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Situation brief and terrain analytics

7.tif | absolute dsm | job site-den

0.647 km² Scene area	480 Buildings detected	8.0m Height above ground (p95)	0.61 m/px Ground resolution
0.53m Mean modelled uncertainty	86.3° Maximum slope	Calibrated Scale calibration	284 of 480 Buildings in flood zone

Narrative written by the offline template analyst from computed scene statistics. Single-view vertical error is metre-scale; surface strength is not assessed. Decision support only, not a survey or an aviation clearance.

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1. Situation report and response plan

Situation report

The scene covers 0.647 km² at 0.61 m per pixel (absolute height model). Surface heights range from 1,676.2 to 1,717.6m; the mean height above ground is 1.8m and the 95th percentile is 8.0m.

480 building footprints were detected. Dominant surfaces: low vegetation (0.193 km²), road (0.193 km²), tree (0.155 km²).

Slopes reach 86.3 degrees (mean 17.8); 0.1580 km² exceeds 30 degrees and 0.1670 km² exceeds 15 degrees.

Ranked risk areas

1. Flooding at a water level of 7.7m above the reference inundates 0.2840 km² (43.9% of the scene) and affects 197 of 480 buildings.
2. The north-east sector is worst hit: 100.0% flooded, mean depth 5.44m.
3. The west sector is worst hit: 78.7% flooded, mean depth 3.66m.
4. The north sector is worst hit: 76.2% flooded, mean depth 1.62m.
5. Steep terrain above 30 degrees (0.1580 km²) is a landslide-prone risk.

Response plan

Landing and relief-drop options (candidate sites for operator confirmation):

- Site 1: clear radius 41.4 m, mean slope 3.7 degrees, score 0.74, 28 m from a road.
- Site 2: clear radius 17.0 m, mean slope 1.9 degrees, score 0.63, 1 m from a road.
- Site 3: clear radius 15.6 m, mean slope 3.2 degrees, score 0.55, 1 m from a road.
- Total flat, obstacle-free candidate area: 0.3250 km². Use the highest-scoring sites outside the flood extent for camps and drops.
- Evacuate the most-flooded sectors first; 197 buildings need priority checks at the modelled level.
- Prefer access routes outside the inundated area; verify roads on the ground before dispatch.

Uncertainty note. Heights come from a single image: vertical error is metre-scale (mean modelled uncertainty 0.53m), surface strength (mud vs tarmac) is not assessed, and landing sites are decision support for a pilot to confirm, not a clearance.

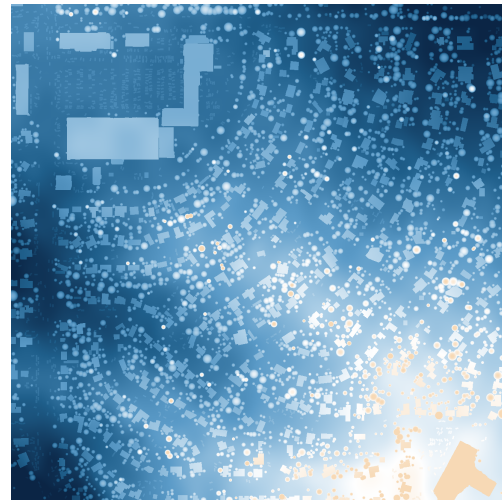
Generated offline by the template analyst from computed statistics; no language model was used.

2. Scene overview

Property	Value
Image	1320 × 1320 px, GeoTIFF
Georeferenced	yes, EPSG:6428
Height model	none, 0 TTA variants, 1 tiles
Scale calibration	calibrated; sources: shadow
Calibration note	Terrain from SRTM 1 arc-second; building and tree heights from measured cast-shadow length at the sun elevation measured in the image.
Sun (shadow geometry)	azimuth 201.6°, elevation 54.6° (measured from cast shadows)
Processing time	33.7 s total, 0.0 s inference



Optical image



Estimated height

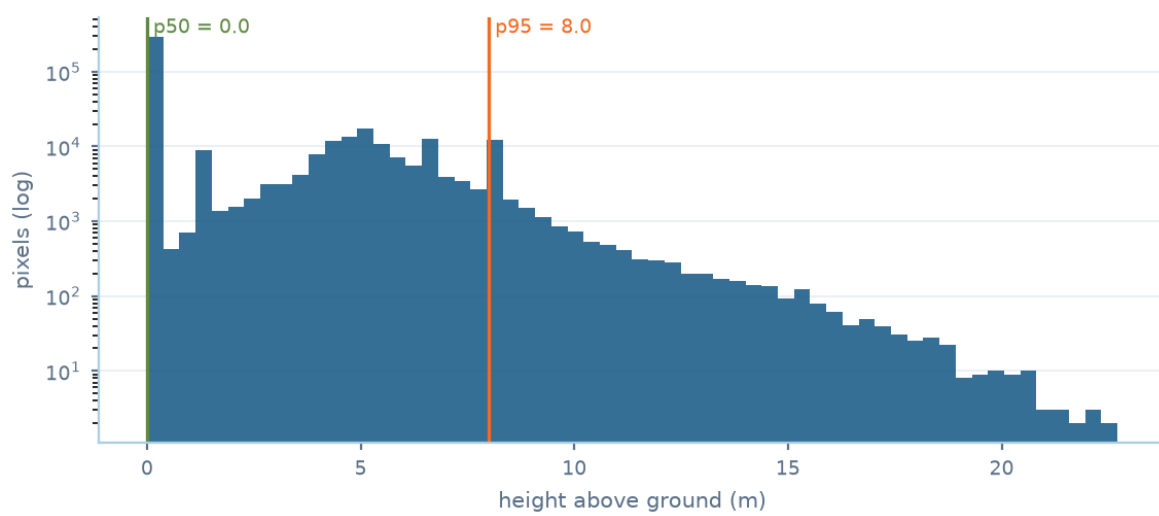


Land cover

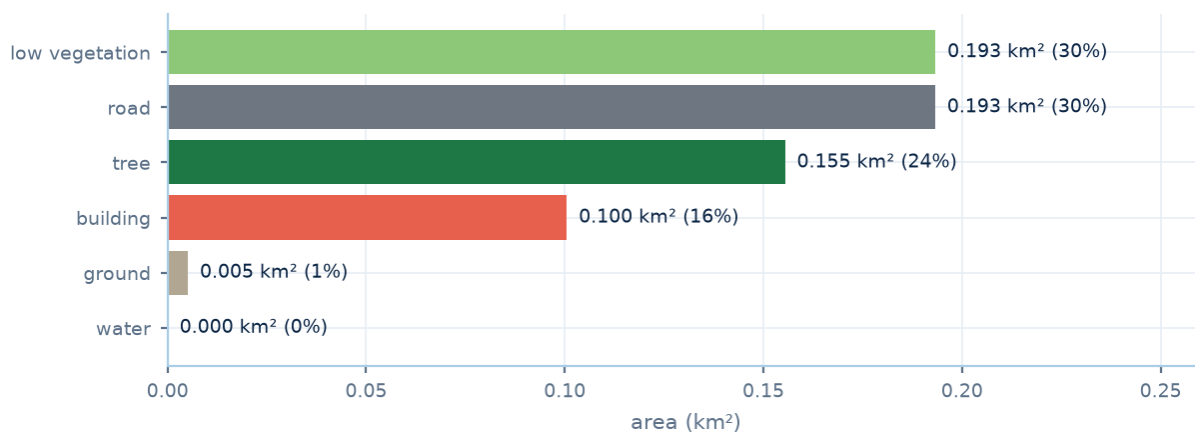


Hillshade

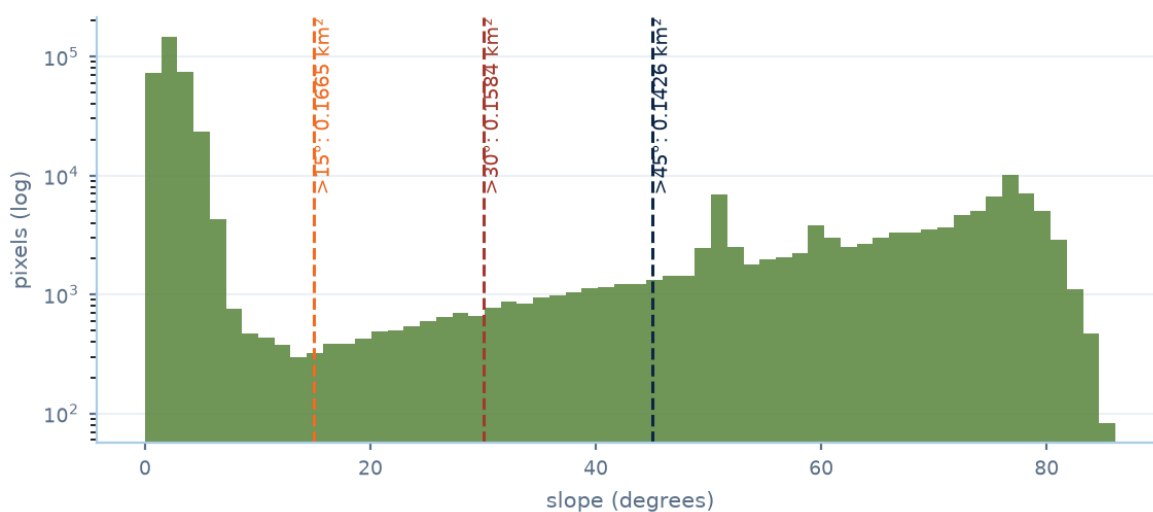
3. Height and terrain analytics



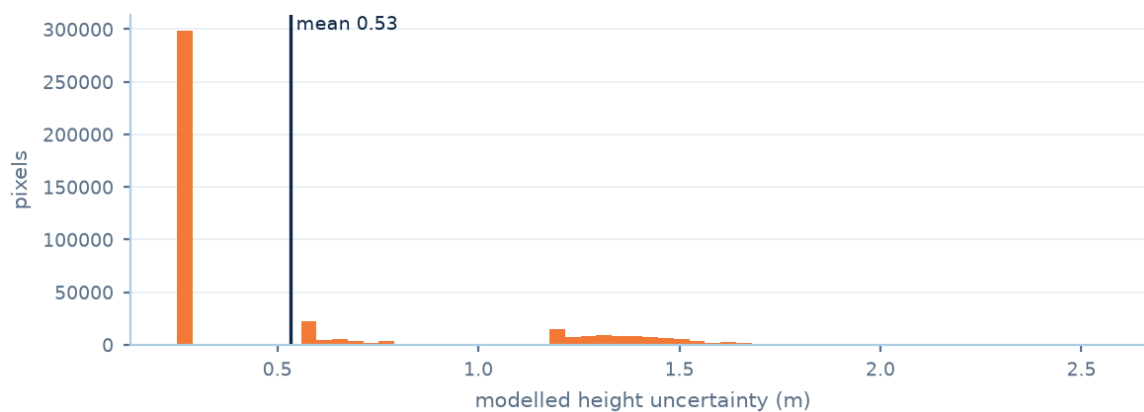
Distribution of height above local ground across the whole scene.



Land-cover composition (colour and height heuristics: indicative, not surveyed).



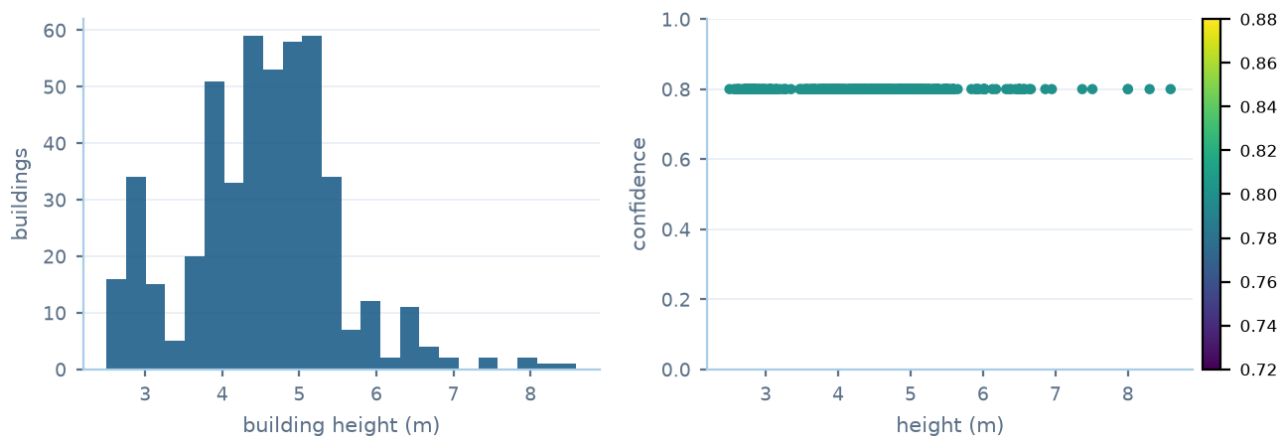
Slope distribution with the area above the 15° , 30° and 45° hazard thresholds.



Per-pixel uncertainty from test-time-augmentation spread (not a measured error).

4. Buildings

481 building footprints with a height estimate. Median 4.6 m, 90th percentile 5.5 m, tallest 8.6 m.

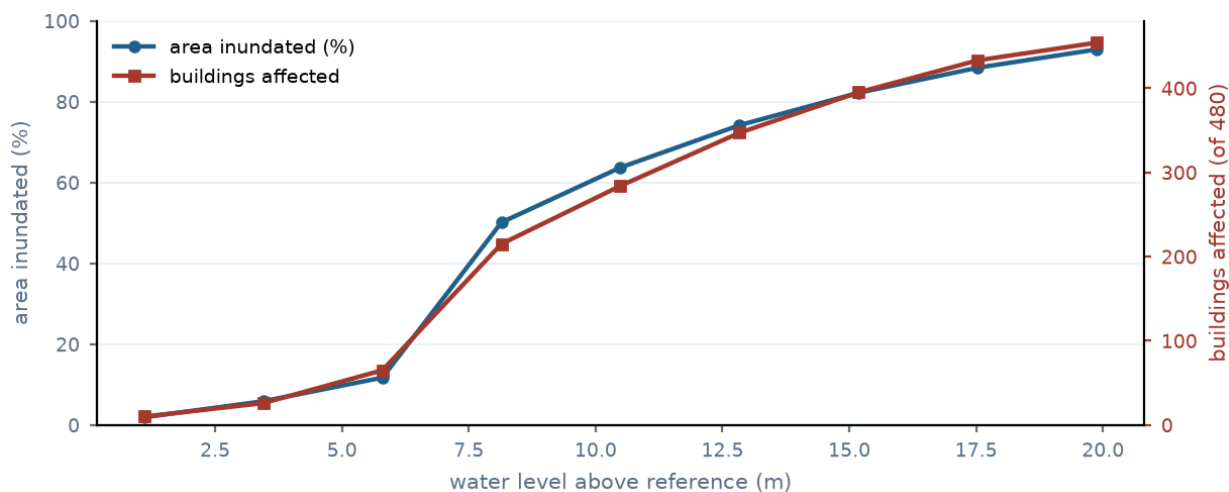


Height distribution (left) and per-building confidence against height (right).

Ten tallest detected buildings

ID	Height (m)	Footprint (px)	Confidence	Shadow check
312	8.6	661	0.80	8.6
370	8.3	262	0.80	8.3
6	8.0	26,796	0.80	8.0
43	8.0	13,237	0.80	8.0
188	7.5	483	0.80	7.5
376	7.4	303	0.80	7.4
402	7.0	442	0.80	7.0
243	6.9	403	0.80	6.9
456	6.7	413	0.80	6.7
3	6.6	2,162	0.80	6.6

5. Flood exposure



Flood response curve: the scene flooded at nine water levels (hydraulically connected to the lowest edge or water body).

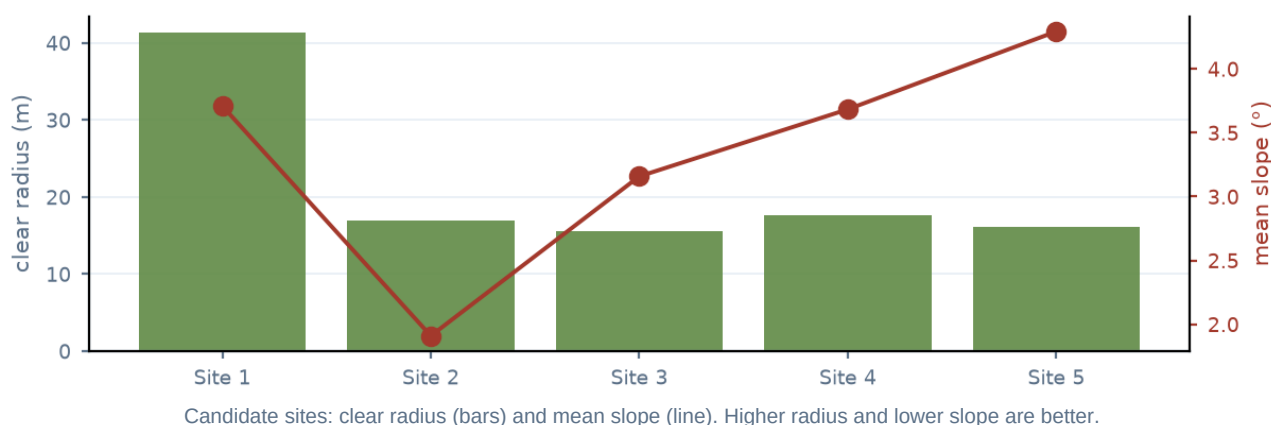
Water level	Area inundated	Buildings affected	Volume
1.1 m	1.9%	10 of 480	14,573 m ³
3.5 m	5.9%	26 of 480	70,330 m ³
5.8 m	11.7%	65 of 480	200,076 m ³
8.1 m	50.1%	215 of 480	1,014,087 m ³

Water level	Area inundated	Buildings affected	Volume
10.5 m	63.7%	284 of 480	1,892,868 m ³
12.8 m	74.2%	347 of 480	2,942,356 m ³
15.2 m	82.2%	395 of 480	4,130,702 m ³
17.5 m	88.4%	433 of 480	5,425,004 m ³
19.9 m	92.9%	454 of 480	6,801,700 m ³

Worst-hit sectors at 10.5 m

Sector	Flooded	Mean depth
north-west	100.0%	2.61 m
north	100.0%	3.85 m
north-east	100.0%	8.20 m

6. Landing and relief-drop sites



Rank	Clear radius	Mean slope	Nearest road	Score
1	41.4 m	3.7°	28 m	0.74
2	17.0 m	1.9°	1 m	0.63
3	15.6 m	3.2°	1 m	0.55
4	17.7 m	3.7°	0 m	0.54
5	16.1 m	4.3°	0 m	0.48

Candidate sites for operator confirmation. Surface strength is not assessed and vertical error is metre-scale: this is decision support, not a landing clearance.

7. Accuracy validation

No reference elevation data has been scored for this scene, so no accuracy figure is claimed. Add a LiDAR DSM/nDSM in the Validation lab to score RMSE, MAE, bias, NMAD, Pearson r and $\delta 1$.

8. Method and limitations

Heights come from a single optical image: a fine-tuned monocular depth model (with test-time augmentation and tiling) produces a height field, which is scaled to metres from shadow geometry, ground control points and/or a reference DEM (see Scene overview for the sources used on this scene).

Limitations: vertical error is metre-scale and larger over dense vegetation and very tall structures; land cover, building footprints and hazard zones are heuristic; flood assumes water spreads only to connected low ground; surface strength (mud, tarmac, load-bearing) is not assessed; landing sites are candidates for a pilot to confirm. This

report is decision support and not a certified survey.